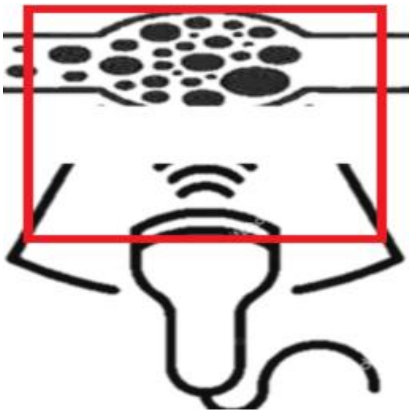
US-DICOMizer



Anonymizing & Labeling of Ultrasound DICOM Files

User Manual for Version 5.2

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1. Installation

Download the latest version released from https://github.com/thrombusplus/US-DICOMizer. Extract the **.exe** file in a local folder of choice. The application does not require installation, run the application by executing the .exe file.

2. Interface components

The main Graphical User Interface (GUI) components are shown in Figure 1:

1| The "**Selected files**" panel.

2| The "**Ordered files**" panel.

3| The "**Image preview**" panel.

4| The "**DICOM Attributes**" panel.

5| The "**Actions**" panel.

6| The "**Anonymized DICOM files**" panel.

7| The "**Menu**".

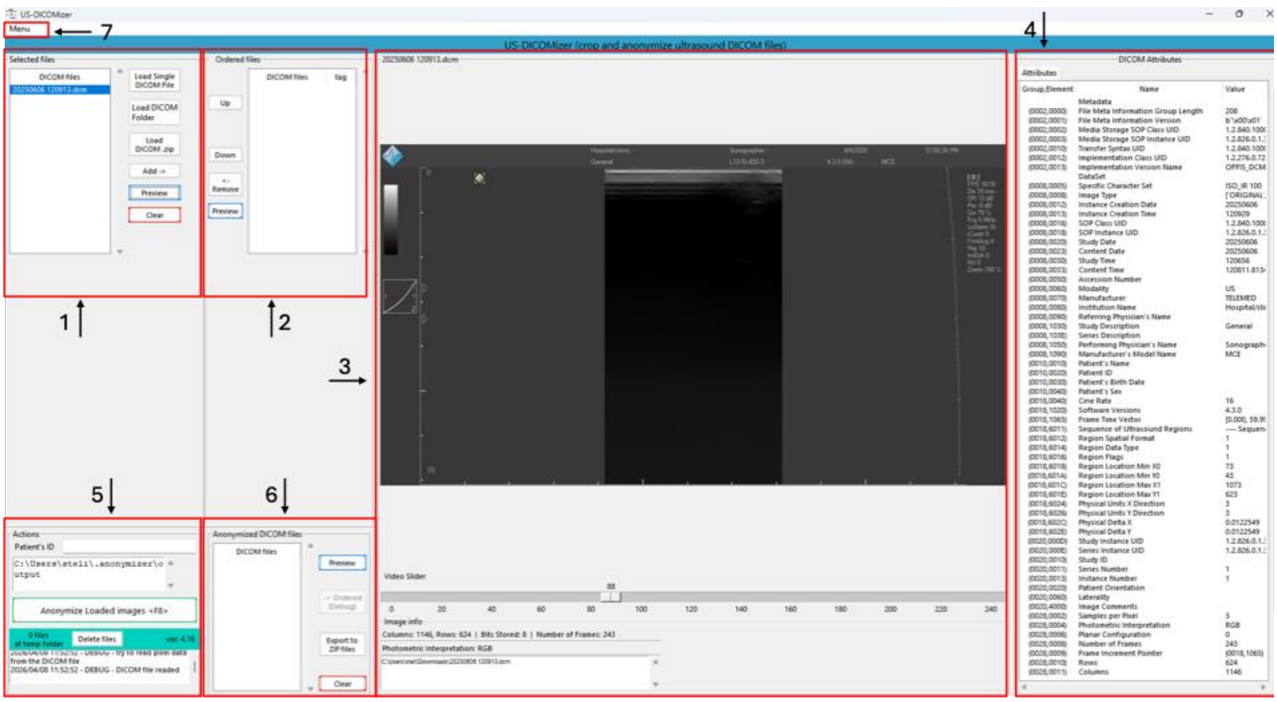

Figure 1. The main application interface components.

3. Load DICOM files

DICOM files can be imported in three different ways, utilizing the respective buttons in the "**Selected files**" panel (Figure 2):

1| **Load Single DICOM File**: Loads a single *.dcm file.

2| **Load DICOM Folder**: Loads multiple *.dcm files by selecting the folder including the files.

3| **Load DICOM .zip**: Loads multiple *.dcm files from a .zip file.

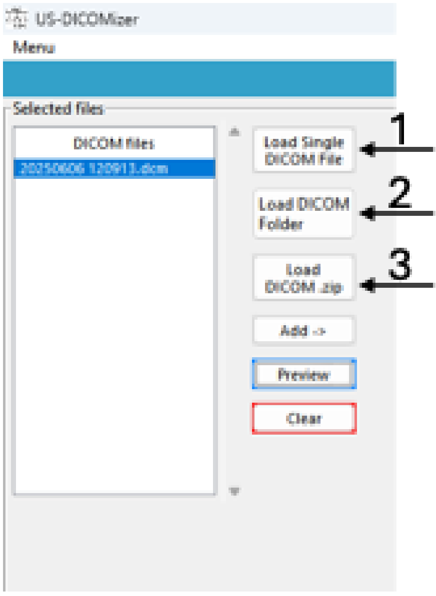

Figure 2. Buttons for loading DICOM files.

In all cases, a new dialog window opens for selection. Based on the above selection, the respective files will be displayed in the “**Selected files**” list by their filenames.

To preview files, select a file and press the “**Preview**” button (or double). The DICOM file will be visualized as image in the “**Image preview**” panel and the respective DICOM attributes will be displayed in the “**DICOM Attributes**” panel (Figure 3).

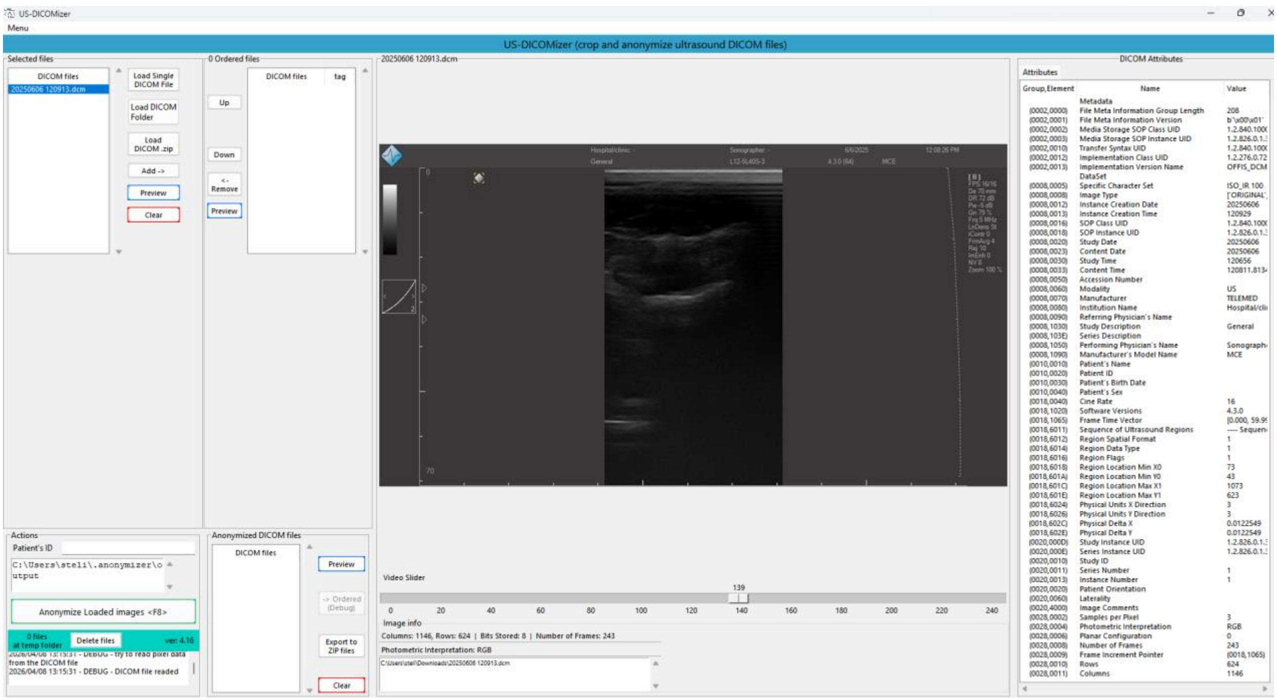

Figure 3. Preview of a DICOM file. Pixel data are visualised in the “Image preview” panel and all DICOM attributes in the respective panel. Image information is displayed below the image, and if the DICOM is a multi-frame file, the sidebar can be used to navigate through different frames.

If a file should be anonymized, press the "**Add ->**" button to move the file to the "**Ordered files**" list. Only files included in the "**Ordered files**" list will be processed and anonymized. Pressing the "**Clear**" button will remove all files from the "**Selected files**" list.

4. Select and order files for anonymization

Only files included in the "**Ordered files**" list will be processed and anonymized. The pane also includes the following four buttons:

- **Up/Down**: Change the order of the selected file.
- **Remove**: Remove the selected file from the "Order files" list and move it back to the "Selected files" list.
- **Preview**: Display the selected file as an image and view its tags.

To preview files, select the file and press the "**Preview**" button (or double click). To remove the file from the list, select the file and press the **"<-Remove**" button. The file will be transferred back to the "Selected files" list. To order files use the "Up" and "Down" buttons, to move the selected file up and down in the list, respectively. Ordering files is important, so files be renamed with an increment number after anonymization accordingly, to match the order of files proposed by the acquisition protocol.

Previewed DICOM files will be visualized in the "**Image preview**" panel and their attributes will be shown in the "**DICOM Attributes**" panel.

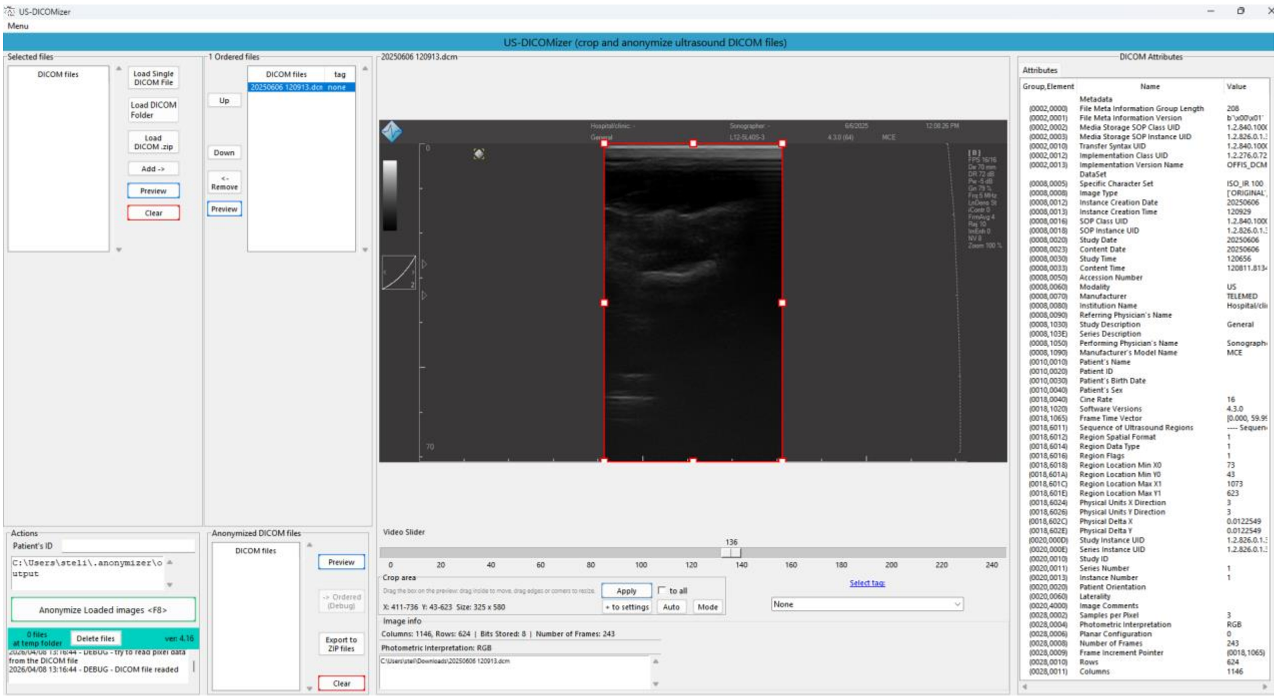

Figure 4. Preview of a DICOM file from the "Ordered list". Pixel data are visualised in the "Image preview" panel and all DICOM attributes in the respective panel. Image information is displayed below the image, and if the DICOM is a multi-frame file, the sidebar can be used to navigate through different frames.

5. Preview images, set cropping area and file tagging

While in preview a red box representing the cropping area will appear in the preview panel. The box can interactively be adjusting by moving dragging the respective edges using the mouse. To facilitate, this process, there are **three** methods for setting the cropping area automatically. These methods are:

- Define cropping box coordinates in the "**Settings**" file.
- **Mode** method.
- **Auto** method.

The **Mode** method uses image statistics and morphological operations to draw the cropping area. The **Auto** method is a hybrid method, that applies multiple processing techniques, including the **Mode** method to find the edges between background and foreground and draw the cropping box. Finaly, cropping coordinates can be adjusted in the settings file. These coordinates are associated with the respective ultrasound machine serial number. If coordinates have been set, the application uses these coordinates to draw the cropping box, every time the file is previewed. Coordinates can be added to the settings, by setting the cropping box appropriately and pressing the **"+ to settings**" button. However, if the same ultrasound machine is used with different scanning depth, resulting in different image dimensions, this method will fail and the other two methods should be utilized.

When the cropping box have been adjusted accordingly, press the "**Apply**" button, to lock these coordinates. This is where the image will be cropped during anonymization. The cropping box will also turn green. If the "**to all**" checkbox is selected before pressing the "**Apply**" button, the same cropping box will be applied to all similar files (single frame or multi-frame) within files in the "**Ordered files**" list.

A tag representing the anatomical site of the ultrasound image can be added to the previewed image using an option from the drop-down menu located below the sliding bar. The selected tag will appear next to the name of the file in the "**Ordered file**" list. If the cropping coordinates have been set, the filename will turn green as well.

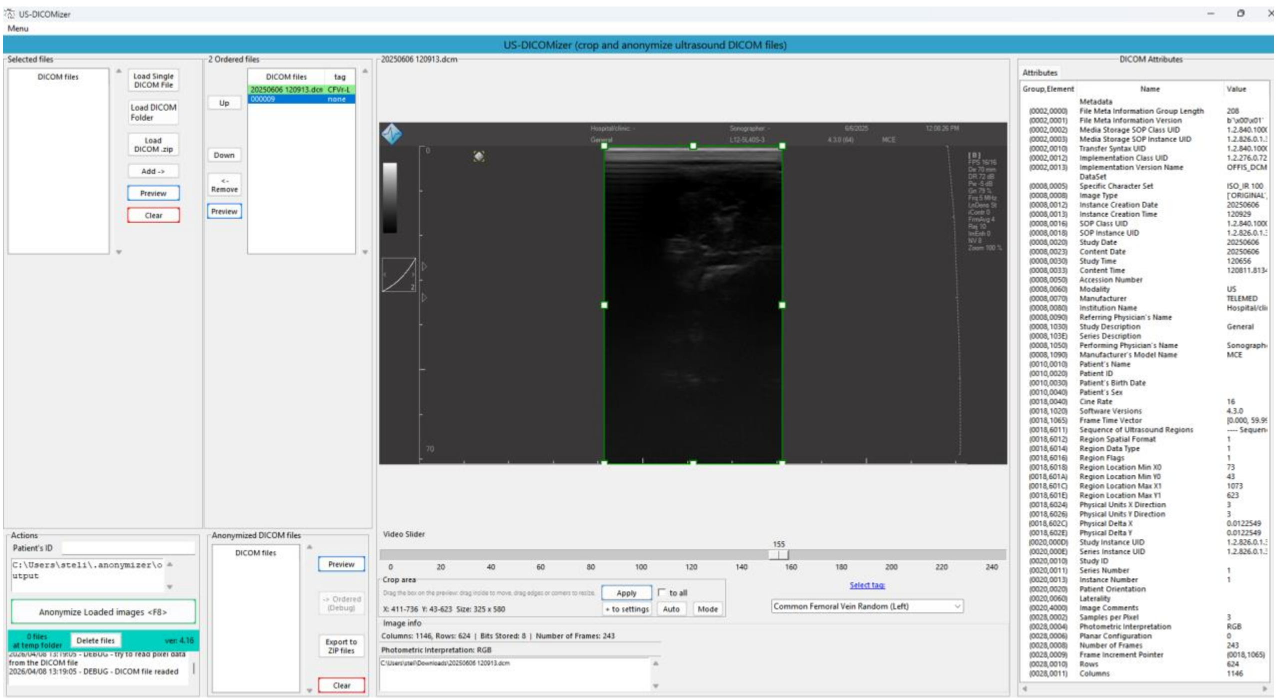

Figure 5. Pressing "Apply" will make the cropping box and the filename in the "Ordered files" list turn green. The respective anatomical tag can also be set using the drop-down menu below the slide bar. The selected tag will also appear next to filename in the "Ordered files" list.

6. Anonymizing

To continue with the anonymization, except the cropping area and the respective tag for all files listed under the “**Ordered files**” list, a proper patient ID should be typed in the “**Patient ID**” filed, under the “**Actions**” panel. Afterwards, by pressing the “**Anonymize Loaded images <F8>**” button will start the anonymization process. All files within the “Ordered list” will be cropped and all attributes containing sensitive information will be removed. The filename of each image will be replaced using the following template: ***anonymized_PATIENTID_X_XXXX_tag.*** The *PATIENTID* is the patient ID typed on the respective field, the X_XXXX represents an incremental number based on the number and order of DICOM files selected and the tag is the tag representing the anatomical site. Anonymized files will be listed in the “**Anonymized DICOM files**” list. To export the anonymized files in a .zip file press the “**Export to .ZIP files”** button. If anonymized files are listed in the “Anonymized DICOM files” list and those files are previewed, then the “**Annotations**” tab in the “**DICOM Attributes**” panel will be activated and be available (Figure 6).

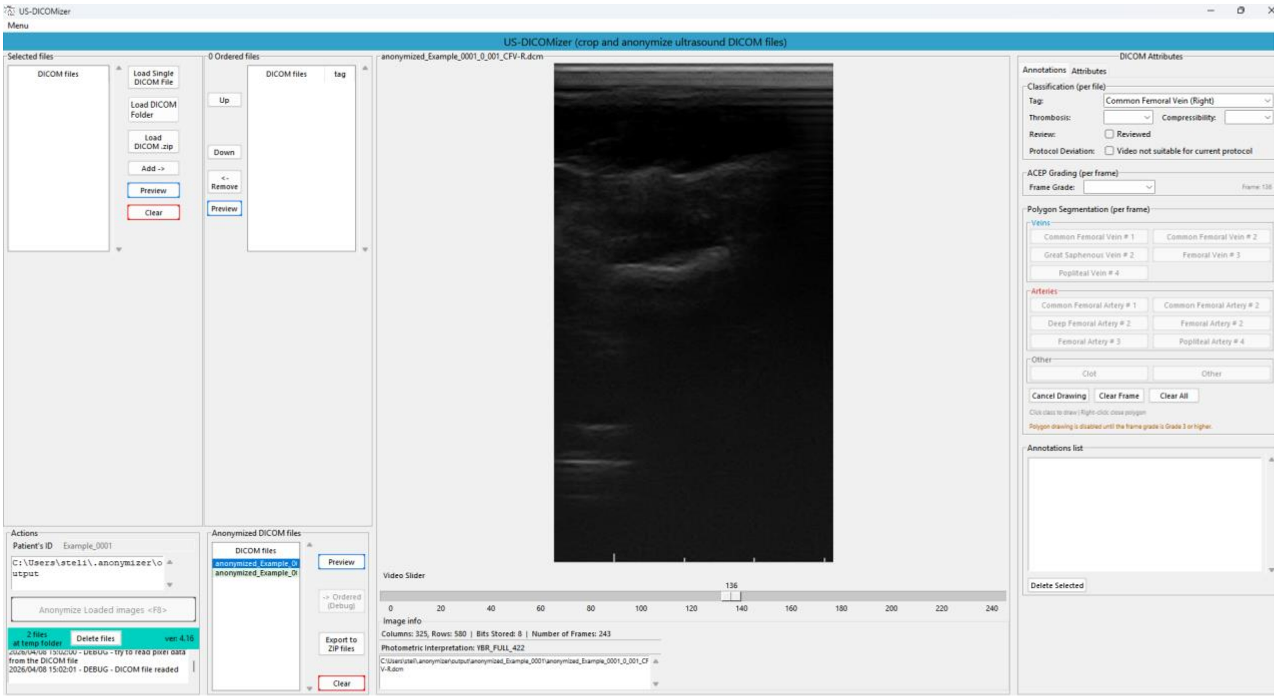

Figure 6. After anonymization process finalized all anonymized files will be listed in the “**Anonymized DICOM files**” list. The “**Annotations**” tab in the “**DICOM Attributes**” panel will be activated.

7. Labeling

For files listed in the “**Anonymized DICOM files**” and be previewed, the “**Annotations**” tab in the “**DICOM Attributes**” panel will be available to perform pixel-wise and file-based labelling. If an .zip file containing already anonymized DICOM files, those files will be listed directly in the “**Anonymized DICOM files**” list, for labelling or editing the already existing labels.

The “**Annotations**” tab contains three panels for labelling and one panel for listing and managing labels. The first panel includes labels which are associated per file, i.e. the anatomical tag selected before anonymization, the presence of Thrombosis and the vein Compressibility. In addition, this panel includes a checkbox for selecting if the annotations of the file has been reviewed and validated and a checkbox for selecting if deviations from acquisition protocol is being observed. If this checkbox is selected and a panel for free typing will made visible. This function is mainly used to describe mainly videos that have been acquired differently than stated in the protocol. The second panel contains the ACEP grading score which is added per DICOM frame. The third panel includes the pixel-wise labels for drawing polygons over the image per frame. The final panel lists all per frames labels for managing them. Adding polygon segmentation labels requires the ACEP grading score label to be set 3 or greater, indicating a frame with good diagnostic value, where anatomical structures are visible and can be identified.

For annotating the anatomical structures, select the respective label from the right panel and add points in the image using left-click. To terminate the process and conclude the polygon use right-click. **Try to draw the anatomical structure with precision, using as few points as possible**. Labels can overlap, or be enclosed within other labels. All added labels drawn will be listed in the "**Annotations list**" panel along their respective frame they have been drawn to. By clicking on the listed labels, the respective frame will be visualized for validation. To delete a selected label, press the "**Delete Selected**" button in the bottom (Figure 7).

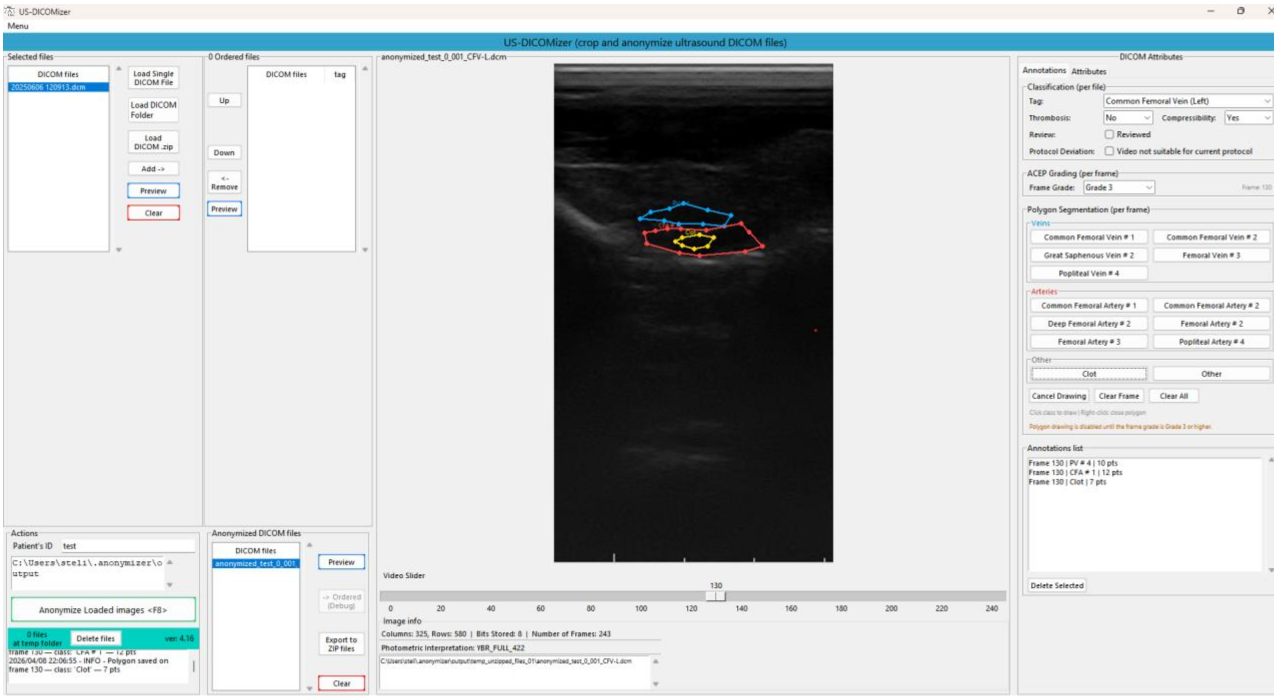

Figure 7. An indicative DICOM frame with three polygon annotations. All polygon annotations are listed in the "Annotations list" alongside their respective frame to which they have been drawn.

8. Export Files

To export the anonymised DICOM files and the respective labels, press the "**Export to ZIP files**" button. A window for selecting the saving location will prompt. The application will create a .zip file with all anonymised DICOM files and a .json file for each DICOM that contains all annotations.

9. Settings and other Functions

From the Menu the Setting window can be accessed. In the settings window, users can control the JPEG compression level applied to images. Values range from 0 (lowest quality) to 95 (highest quality). Values above 95 are considered uncompressed. A value of 75 is recommended. Additionally, users can change the folder where temporary files are extracted and the format of the .json file thay stores all the annotations. By default the structure is similar to the .json file exported from the Darwin V7 labs platform. If the US-DICOMizer runs in a personal computer, it is highly advised to fill in the **Clinician Name**. This name will be also included in the annotations file for future reference. In the bottom, the Serial Number (SN) and the predefined cropping coordinates for these ultrasound machines are shown. Pressing the "**Settings folder**" button will open the folder where the **Settings.ini** file is stored. Re-starting the application is required if modifications to the **Settings.ini** file has been performed.

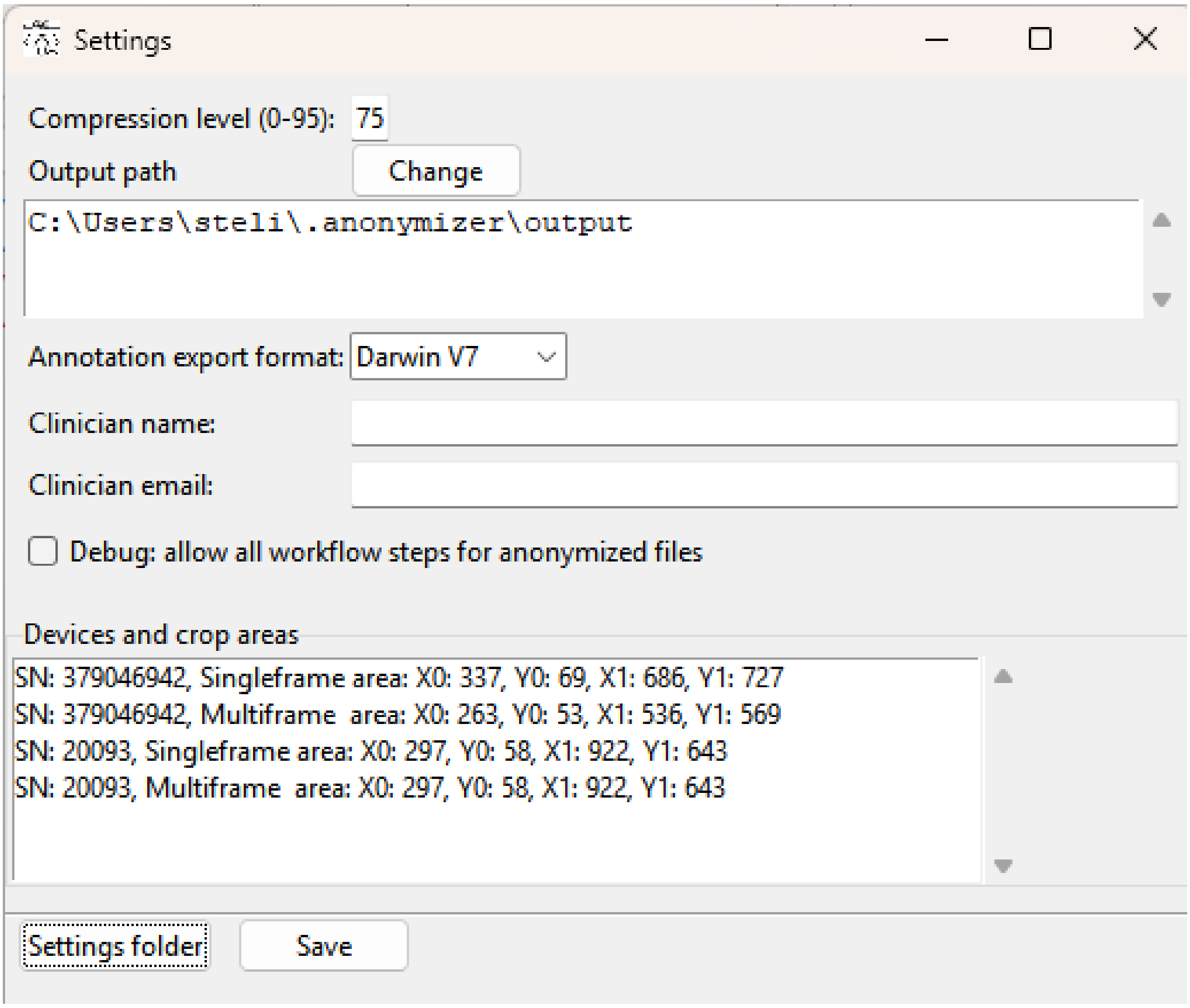

Figure 8. The settings window.

10. Keyboard Shortcuts

Keyboard shortcuts replace button actions or execute additional processes.

- **F6** – Transfers all files from the Selected files list to the Ordered files list
- **F8** – Executes anonymization, replacing the Anonymize Loaded images button
- **F9** – Replaces the Export to zip folder button
- **PG UP / PG DN** – The Page Up and Page Down keys on your keyboard execute the
- **Up / Down** commands to change the order of the selected file.